

Scalar Magnetic Potential Interpolation for Non-Conformal Meshing in Mesh-Based Generated Reluctance Networks

S. Asfirane^{1,2}, S. Hlioui³, S. Mezani⁴, Y. Amara², O. De La Barriere¹, G. Barakat², and M. Gabsi¹

¹SATIE, CNRS, ENS Paris-Saclay, 94235 Cachan, France

²GREAH, Université Le Havre Normandie, 76600 Le Havre, France

³SATIE, CNRS, CNAM, F-75141 Paris, France

⁴GREEN, Université de Lorraine, 54000 Nancy, France

The aim of this paper is to assess the efficacy of non-conformal meshing in mesh-based generated reluctance network modeling. A representative semi-numerical example is introduced to demonstrate the possibilities and accuracy offered by a non-conformal meshing. In a conformal mesh, two reluctance block elements share the same branch. This makes the computation faster and more accurate but is not convenient for motion processing or mesh relaxation and it does not solve the air-gap modeling problem (rotor/stator connectivity). To solve this, different approaches are used. The interpolation approach developed in this paper aims to render motion processing independent of discretization. This approach is also applied to couple different meshes (spatial discretization) and makes it possible to connect a lumped parameter model to a meshed one. The model is divided into independent zones that are connected via the interpolation coupling. Several reluctance network (RN) non-conformal meshes are presented and global quantities are compared to a fine-meshed finite-element model. Magnetic saturation is considered in the iron parts. Comparisons are provided under open-circuit and on-load configurations for flux linkage and force. The main advantage is to overcome the limitations of generic RN for which the movement modeling requires remeshing at each position. The drawbacks of the method are the increased number of variables (additional interface nodes) and a certain loss of accuracy due to the interpolation function order.

Index Terms—Magnetic equivalent circuits (MECs), mesh relaxation, non-conformal mesh, permanent magnet (PM), reluctance network (RN).

I. INTRODUCTION

THIS contribution is a part of a wider project aiming to develop a magnetic field computation tool which uses a mesh-based generated reluctance networks (MGRNs) technique [1], [2]. Magnetic equivalent circuits (MECs) have regained popularity among machine designers but still suffer from the lack of genericity. The purpose of the tool and the developed techniques is the standardization of reluctance networks (RNs) to manage different aspects of an electrical machine modeling problem. Some computer-aided design software using RN modeling exists. Turbo-TCM [3] was developed for the modeling of small power alternators. RMxprt in the ANSYS electromagnetic [4] package features predefined topologies of stators and rotors. SPEED [5] is another tool based on analytical formulations complementary to the finite-element analysis (FEA). Reluctool [6] is based on lumped parameter MECs for which the RN requires designer's expertise. The aim of the MGRN approach is to free the designer from building a specific model for a given topology and to allow the automatic processing from the mesh generation up to equations extraction and matrix solving. In this paper, to enrich this method and to contribute to the development of techniques allowing the standardization of the procedures, a special focus is made on the automatic meshing aspect. A representative

semi-numerical example is introduced to demonstrate the possibilities and accuracy of non-conformal meshing in MGRN. The non-conformal mesh is used in some FEA commercial software for motion consideration as to avoid remeshing at each time step in transient analysis. In a conformal RN mesh, two reluctance block elements share the same branch. This makes computation faster and more accurate but as it remains dependent on discretization, it is not convenient for motion processing. In addition, it does not solve the air-gap modeling problem when the rotor and stator are meshed independently, and then connected on a non-conformal interface. To solve this latter problem, different approaches are used in the literature. Hybrid analytical models [7] are used to compute the magnetic field solution in constant permeability regions [air gap and permanent magnet (PM)] via an analytical formal solution, while the field solution in the slots and the ferromagnetic parts is evaluated through RN modeling. In [8], variable RN is used. The air gap is re-meshed at each step inducing a change in the size of matrix system. Additional tangential reluctances in the air gap are used in [9] but it is limited to air gap and would be difficult to implement for mesh relaxation. Interpolation coupling is used in [10] to connect the stator and rotor systems and to render motion processing independent of discretization. In this paper, the latter approach is developed and applied to couple different meshes (spatial discretizations). It also allows applying a mesh relaxation on different zones of the model that does not need the same mesh density (back iron as opposed to air gap). The main advantage of this technique is the fact that it allows to improve the genericity of the developed tool. On the plus side, this method helps overcome

Manuscript received November 5, 2018; revised January 29, 2019; accepted February 10, 2019. Date of publication April 12, 2019; date of current version June 20, 2019. Corresponding author: S. Asfirane (e-mail: salim.asfirane@gmail.com).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TMAG.2019.2899820

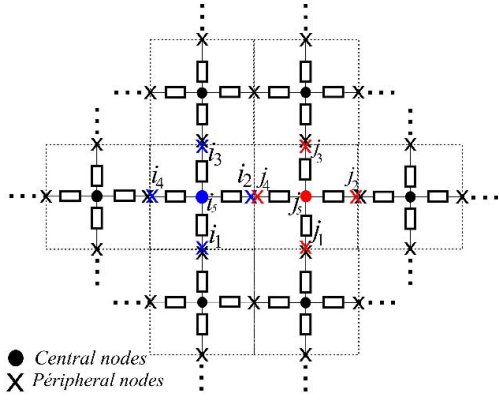


Fig. 1. Conformal mesh reluctance block elements.

one of the limitations of classical RN modeling, related to the consideration of motion. The goal is to use the Lagrange interpolation method to allow for flexibility in the mesh of the field of study, while preserving the sparsity of the resolution matrix. The drawbacks of the method are the increased number of variables (additional interface nodes) and a certain loss of accuracy due to the interpolation function order.

II. AUTOMATIC MESH-BASED GENERATED RELUCTANCE NETWORK

Some operations are carried out to set the matrix system to solve. These operations include the automatic creation of the RN (spatial discretization), the computation of reluctances and sources, and setting the matrix equations. In a nodal-based scalar magnetic potential formulation, the matrix equation system is given by

$$[U]_{n \times 1} = [P]_{n \times n}^{-1} \times [\phi s]_{n \times 1} \quad (1)$$

where $[P]$, $[U]$, and $[\phi s]$ are the permeance matrix, the unknown magnetic potential vector, and the flux source vector, respectively. n represents the total number of nodes.

A. Node/Reluctance Block Element

It is important to make the distinction between a reluctance block element and a node. In a conformal mesh, central nodes of block elements are connected via their shared branches. The reluctance blocks elements, when taken two by two, share the same branch. It becomes easy to sum both reluctances of the branch to perform a connection between both central nodes. In such a system, the number of nodes is the same as the number of elements and each node is unique to each block element. In this paper, we are going to differentiate the central nodes from the peripheral nodes. Fig. 1 illustrates a conformal RN mesh. The block elements i and j are the bordered with peripheral nodes numbered from 1 to 4. The so-called peripheral nodes will disappear from the equation system at the equation-building step with the connection of the block elements.

Fig. 2 represents the studied example that consists of one PM pole excited by a single coil. The dimensions of the model have been exaggerated for clarity reasons. Geometric, electrical, and magnetic parameters of the model are given in Table I. $J_{\max}$ and S_{slot} are, respectively, the current density

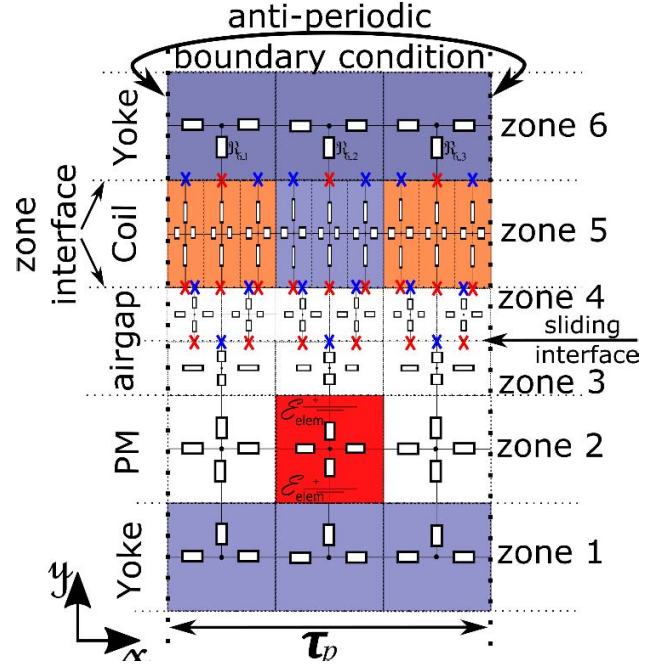


Fig. 2. MGRN model discretization.

TABLE I
MODEL PARAMETERS

Parameter	Value
air-gap, yoke's width, τ_p	10, 10, 30 mm
PM magnetic remanence	1.1 T
Coil current	$(J_{\max} S_{\text{slot}}) \sin\left(\frac{\pi}{\tau_p} x_d\right)$
$J_{\max}, S_{\text{slot}}$	50 A/mm ² , 100 mm ²

and slot cross section. τ_p is the pole width and x_d is the displacement. Current density is set to a high value with the purpose of testing the robustness of the method even in a highly saturated case.

From the model in Fig. 2, it can be seen that zones 1–3 consist of conformal mesh, while zones 4–6 have a non-conformal mesh. The air gap includes a sliding interface between zones 3 and 4 to allow for motion between the moving and the stationary parts. To ensure scalar magnetic potential and magnetic flux equality at the non-conformal interfaces, a first-order interpolation is used (zones 4–6).

The “x” marks indicate additional nodes to be incorporated in the matrix system of the electromagnetic problem. The scalar magnetic potentials at the interface nodes need to appear explicitly in the set of equations to ensure the balance between the number of equations and the number of unknowns in the algebraic equation system (see Section III).

B. Automatic Generation of Matrix System Equations

At the meshing step, the reluctance block elements are created and the geometry is divided into zones, the sources are computed at each displacement step and distributed along the y-axis branches of the block elements. PM sources are placed in the direction of magnetization.

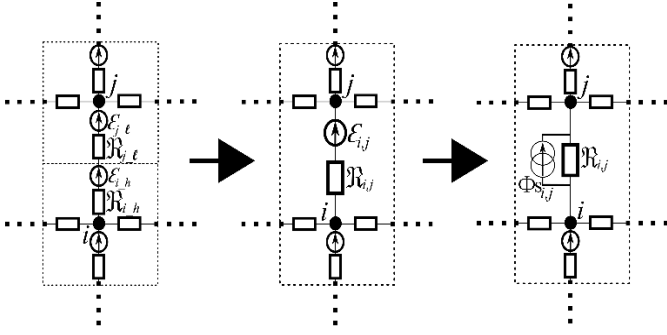


Fig. 3. Connection between adjacent blocks in a conformal mesh-based RN.

Fig. 3 illustrates the connection between adjacent blocks in a conformal mesh-based RN.

$\mathcal{R}_{i,h}$, $\mathcal{R}_{j,l}$, $\varepsilon_{i,h}$, and $\varepsilon_{j,l}$ are, respectively, the reluctances and magnetomotive force (MMF) of nodes i and j .

$\mathcal{R}_{i,j}$, $\varepsilon_{i,j}$, and $\phi s_{i,j}$ are, respectively, the reluctance, the total MMF source, and the flux source on the branch linking the nodes i and j .

Fig. 4 shows the coils current MMF sources distribution where N_s is the number of turns and I is the value of current. More details on MGRN conformal mesh modeling regarding spatial discretization, boundary conditions, and MMF sources modeling and distribution can be found in [2] and [11]. An important issue concerns the numbering pattern of elements since it affects the sparsity of the matrix. Equation (2) shows the developed resolution matrix system for the conformal MGRN model divided into six zones. The saturation of the iron parts (yokes and tooth) is taken into account in an iterative procedure [2]. The permeability values of ferromagnetic blocks are readjusted at each iteration until convergence

$$\begin{bmatrix} [P_{z1}] & [P_{z1-z2}] & [0] & [0] & [0] & [0] \\ [P_{z2-z1}] & [P_{z2}] & [P_{z2-z3}] & [0] & [0] & [0] \\ [0] & [P_{z3-z2}] & [P_{z3}] & [P_{z3-z4}] & [0] & [0] \\ [0] & [0] & [P_{z4-z3}] & [P_{z4}] & [P_{z4-z5}] & [0] \\ [0] & [0] & [0] & [P_{z5-z4}] & [P_{z5}] & [P_{z5-z6}] \\ [0] & [0] & [0] & [0] & [P_{z6-z5}] & [P_{z6}] \end{bmatrix} \times \begin{bmatrix} [U_{z1}] \\ [U_{z2}] \\ [U_{z3}] \\ [U_{z4}] \\ [U_{z5}] \\ [U_{z6}] \end{bmatrix} = \begin{bmatrix} [\phi s_{z1}] \\ [\phi s_{z2}] \\ [\phi s_{z3}] \\ [\phi s_{z4}] \\ [\phi s_{z5}] \\ [\phi s_{z6}] \end{bmatrix} \quad (2)$$

where $[P_{zi}]$ is the zone permeance matrices describing all the connection between the nodes inside the zone i .

$[P_{zi-j}]$ is the coupling matrices between the zones i and j with $i \neq j$ and $i, j = 1, 2, 3, \dots, n_z$.

n_z is the number of zones.

All these matrices are square and their dimension depends on the number of nodes, which is equal to the number of elements in a conformal RN mesh.

$[U_{zi}]$ is the scalar potential vector of the central nodes of the reluctance block elements of zone i . They represent the unknowns in the equation.

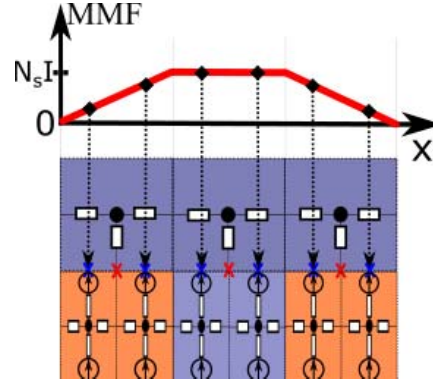


Fig. 4. Coil magnetomotive sources distribution.

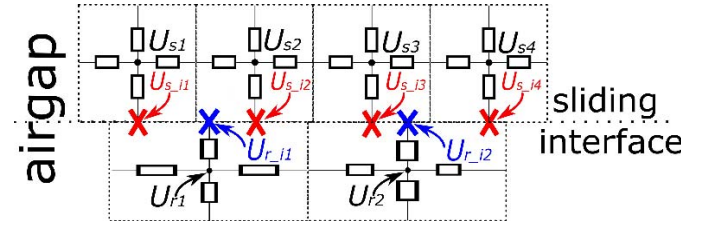


Fig. 5. Sliding surface block elements and interface nodes.

$[\phi s_{zi}]$ is the source vector attached to nodes of zone i expressed as a flux source to fulfill the homogeneity of the equation.

C. Global Quantities Computation

Flux linkage is computed as the sum of branch fluxes (in the y -axis) passing through all block elements located at the slot mid-height spanning one pole pitch (see zone 5 on Fig. 2). Electromotive force (EMF) is obtained by differentiation of flux linkage. Force computation is based on Maxwell stress tensor and is calculated by (3). F is the 'force, l_{i-x} , B_{i-x} , and B_{i-y} are the x -axis length and the x - and y -axes flux density values of air-gap block element i , respectively. μ_0 is the vacuum permeability and L_a is the active length.

$$F(x_d) = \frac{1}{\mu_0} L_a \sum_i l_{i-x} B_{i-x}(x_d) B_{i-y}(x_d). \quad (3)$$

III. AUTOMATIC MGRN WITH MAGNETIC SCALAR POTENTIAL INTERPOLATION

A. Air-Gap Modeling and Motion Processing

Based on the continuity of scalar magnetic potential and the continuity of magnetic flux at the sliding interface, the first-order linear interpolation is used to evaluate the scalar magnetic potential values at the interface nodes.

Fig. 5 illustrates a non-conformal mesh at the air gap. The magnetic scalar potential of block elements central nodes is labeled U_r and U_s for the moving and stationary zones, respectively. The block element peripheral nodes at the interface are labeled $U_{r,i}$ and $U_{s,i}$. The continuity of the scalar magnetic potential at the sliding interface allows writing each potential as a function of the nodes at its sides. The interface nodes of the moving part will be written as a function of the

scalar magnetic potential of interface nodes of the stationary part. The coupling between the interface nodes of both zones is performed by use of the first-order Lagrange interpolation as shown in (4). The continuity of the magnetic flux density will be ensured by the same interpolation method. Magnetic flux density of interface nodes of the stationary part will be written as a function of the flux density of interface nodes of the moving part as shown by (5). x_{r_i1} , x_{r_i2} , x_{s_i1} , x_{s_i2} , U_{r_i1} , U_{r_i2} , U_{s_i1} , U_{s_i2} , B_{r_i1} , B_{r_i2} , and B_{s_i2} are the positions, scalar magnetic potential, and flux density values of interface nodes of the moving and the stationary part, respectively. When rewritten as a gradient of scalar magnetic potential to meet the nodal formulation, (5) becomes (6). $\mathfrak{R}_{r_i1}$, S_{r_i1} , $\mathfrak{R}_{r_i2}$, S_{r_i2} , $\mathfrak{R}_{s_i2}$, and S_{s_i2} are the reluctances linking the interface nodes to the central nodes of the moving and the stationary part and their cross sections, respectively,

$$U_{r_i1} = \frac{x_{r_i1} - x_{s_i1}}{x_{s_i2} - x_{s_i1}} U_{s_i2} + \frac{x_{s_i2} - x_{r_i1}}{x_{s_i2} - x_{s_i1}} U_{s_i1} \quad (4)$$

$$B_{s_i2} = \frac{x_{s_i2} - x_{r_i1}}{x_{r_i2} - x_{r_i1}} B_{r_i2} + \frac{x_{s_i2} - x_{r_i2}}{x_{r_i2} - x_{r_i1}} B_{r_i1} \quad (5)$$

$$\frac{\Delta U_{s_i2}}{\mathfrak{R}_{s_i2} S_{s_i2}} = \frac{x_{s_i2} - x_{r_i1}}{x_{r_i2} - x_{r_i1}} \frac{\Delta U_{r_i2}}{\mathfrak{R}_{r_i2} S_{r_i2}} + \frac{x_{s_i2} - x_{r_i2}}{x_{r_i2} - x_{r_i1}} \frac{\Delta U_{r_i1}}{\mathfrak{R}_{r_i1} S_{r_i1}}. \quad (6)$$

When generalized to all interface nodes, (4) and (6) will be written in matrix form as shown in (7) and (8).

$[I]$ is the identity matrix. $[U_{int_r}]$ and $[U_{int_s}]$ are the vectors containing the interface nodes scalar magnetic potential values of the moving and stationary parts, respectively.

$[U_r]$ and $[U_s]$ are the vectors of the central nodes scalar magnetic potential values to which the interface nodes are connected. $[M_{interp_s}]$ is the interpolation matrix. $[M_{ps_s}]$ is the matrix containing the permeance of the element divided by its cross section.

$[M_{interp_r}]$ is the interpolation matrix divided by the product of the reluctance of the block and its cross section. Incorporating (7) and (8) in the matrix system will provide the missing equations to balance the number of unknowns in the whole system

$$[I][U_{int_r}] - [M_{interp_s}][U_{int_s}] = 0 \quad (7)$$

$$[M_{ps_s}][U_{int_s}] - [M_{ps_s}][U_s] - [M_{interp_r}][U_{int_r}] + [M_{interp_r}][U_r] = 0. \quad (8)$$

It implies that modifications need to be made in the resolution matrix presented in (2). The scalar magnetic potential

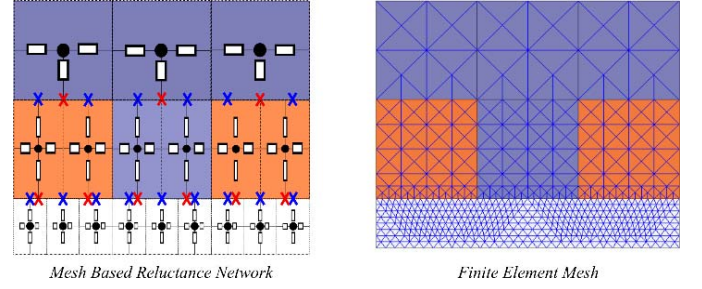


Fig. 6. Mesh relaxation in MGRN and in FEA.

interpolation technique will induce small changes without disturbing the sparsity of the matrix.

In a strong coupling approach, the resolution matrix for the six zone model (see Fig. 2) with one non-conformal interface at the air gap is given by (9), as shown at the bottom of this page. $[P_{part_i}]$ is the permeance linking the interface node to the central node of the same block in zone number i . $[U_{int_i}]$ is the scalar magnetic potential vectors of interface nodes. The rest of the sources array will be filled with zeros since no sources are attached to the interface nodes on both sides (zones). Motion will be incorporated in an easy fashion since the only change that will occur is about positions of interface nodes of the moving part.

B. Mesh Relaxation

In 2-D FEA, mesh relaxation can be done naturally due to the shape of the elements and the formulation specific to the finite-element method. In meshed RN, the spatial discretization is bound by the flux tube principle. Depending on discretization, scalar potentials of interface nodes need to be evaluated. Fig. 6 illustrates the mesh relaxation of MGRN and FEA models. Based on the same approach used to model the non-conformal interface at the air gap (given in Section III-A), the Lagrange interpolation method is used to have a flexibility in the mesh of the field of study.

When applied to different zones of the model, coupling equations need to be created to complete the matrix system. If there is more than one non-conformal interface, the number of variables will increase (additional interface nodes) and the matrix system described in (9) will include more equations. The comparative study on different mesh patterns with mesh relaxation will be detailed in Section IV.

IV. COMPARATIVE STUDY AND MESH ANALYSIS

For the comparative study, the FEA model is sufficiently meshed to provide an accurate reference for comparison.

$$\begin{bmatrix} [P_{z1}] & [P_{z1-z2}] & [0] & [0] & [0] & [0] & [0] & [0] \\ [P_{z2-z1}] & [P_{z2}] & [P_{z2-z3}] & [0] & [0] & [0] & [0] & [0] \\ [0] & [P_{z3-z2}] & [P_{z3}] & [0] & [0] & [0] & [P_{part3}] & [0] \\ [0] & [0] & [0] & [P_{z4}] & [P_{z4-z5}] & [0] & [0] & [P_{part4}] \\ [0] & [0] & [0] & [P_{z5-z4}] & [P_{z5}] & [P_{z5-z6}] & [0] & [0] \\ [0] & [0] & [0] & [0] & [P_{z6-z5}] & [P_{z6}] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] & [0] & [I] & [M_{interp_3}] \\ [0] & [0] & [M_{ps_3}] & [M_{interp_4}] & [0] & [0] & [M_{ps_3}] & [M_{interp_4}] \end{bmatrix} \times \begin{bmatrix} [U_{z1}] \\ [U_{z2}] \\ [U_{z3}] \\ [U_{z4}] \\ [U_{z5}] \\ [U_{z6}] \\ [U_{int3}] \\ [U_{int4}] \end{bmatrix} = \begin{bmatrix} [\phi_{s_{z1}}] \\ [\phi_{s_{z2}}] \\ [\phi_{s_{z3}}] \\ [\phi_{s_{z4}}] \\ [\phi_{s_{z5}}] \\ [\phi_{s_{z6}}] \\ [0] \\ [0] \end{bmatrix} \quad (9)$$

TABLE II
MESH SCHEMES

Zone div.		Mesh schemes									
		Mesh 1	Mesh 2	Mesh 3	Mesh 4	Mesh 5	Mesh 6	Mesh 7	Mesh 8	Mesh 9	Mesh Conf.
Zone 1	x	33	33	21	15	15	9	45	15	45	60
	y	7	7	5	3	1	1	11	7	11	9
Zone 2	x	33	33	21	15	15	9	45	15	45	60
	y	7	7	5	3	3	3	11	7	11	9
Zone 3	x	33	33	21	15	15	9	45	15	45	60
	y	7	7	5	3	3	3	11	7	11	9
Zone 4	x	45	45	27	27	21	12	33	51	51	60
	y	7	7	5	3	3	3	11	7	11	9
Zone 5	x	45	45	36	16	10	10	33	51	51	60
	y	7	7	5	3	3	3	11	7	11	9
Zone 6	x	45	15	15	6	6	9	33	51	51	60
	y	7	7	3	1	1	1	11	7	11	9
Number of blocks		1638	1428	675	270	207	138	2574	1386	3168	3240

Ten MGRN mesh patterns are tested and compared to FEA. Nine of which are non-conformal MGRN patterns used to evaluate the precision of the scalar interpolation approach. A tenth MGRN mesh scheme is applied in a conformal pattern to provide some insight on the pertinence of the scalar interpolation approach with regards to precision and computation time. Table II summarizes the spatial discretization in each zone for each mesh pattern. The number of divisions in the x -axis and y -axis is listed along with the total number of elements. Each mesh pattern has been chosen according to certain specifications. Mesh 1 is a relatively fine mesh with non-conformity on the air-gap interface. Mesh 2 is the same as mesh 1 but with mesh relaxation on the yoke (zone 6). Mesh 3 features a different division pattern with relaxation on zones 5 and 6. Mesh 4 and Mesh 5 have the same pattern as Mesh 3 with fewer elements in the teeth (zone 5). Mesh 6 has the fewest number of divisions. Mesh 7 features a higher number of the x -axis divisions in zones 1–3 than in 4–6. In Mesh 8, a larger difference between the number of the x -axis divisions between zones 1–3 and 4–6 is introduced. Mesh 9 is a dense mesh having a number of nodes close to FEA. “Mesh_Conf.” is a conformal mesh scheme with the same block division number in each zone of the model.

Figs. 7 and 8 show the open-circuit flux linkage turn versus displacement and its harmonic content, respectively. Figs. 9 and 10 show, respectively, the back EMF per turn and harmonic content for all mesh patterns. The same harmonic content distribution found in flux linkage waveforms will also be found in back EMF waveforms as the latter is calculated via the flux linkage derivative. On most harmonics, the highest magnitude difference when compared to FEA waveforms are found on mesh patterns 3 and 6, especially on 11 and 13 harmonics. The last mentioned mesh patterns contain a non-conformal interface on three levels (zones 3 and 4; 4 and 5; and 5 and 6; see Fig. 2 and Table II). The same observation is made on mesh number 5. The interpolation coupling is used on all three levels inducing more discrepancies on harmonic content than other mesh patterns with less non-conformal interfaces

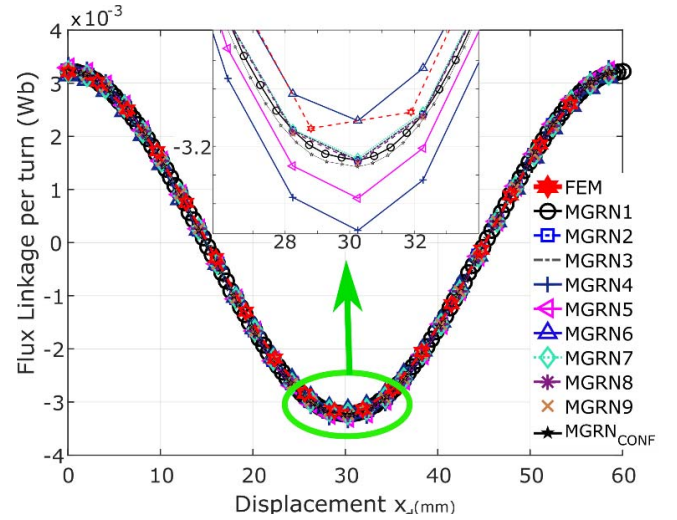


Fig. 7. Open-circuit flux linkage per turn versus displacement.

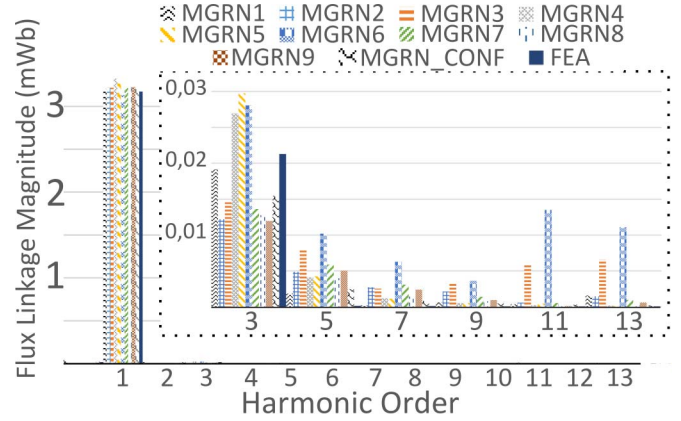


Fig. 8. Open-circuit flux linkage per turn harmonic content.

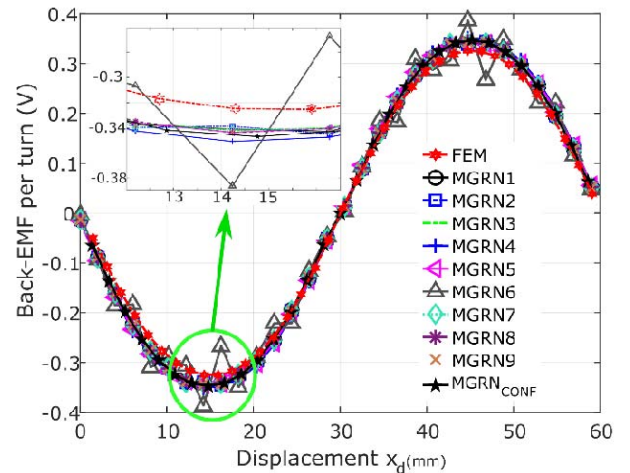


Fig. 9. Open-circuit back EMF per turn versus displacement.

(one interface for meshes number 1, 8, and 9 or two interfaces for mesh number 2).

Figs. 11 and 12 show the cogging force waveforms versus displacement and harmonic content, respectively.

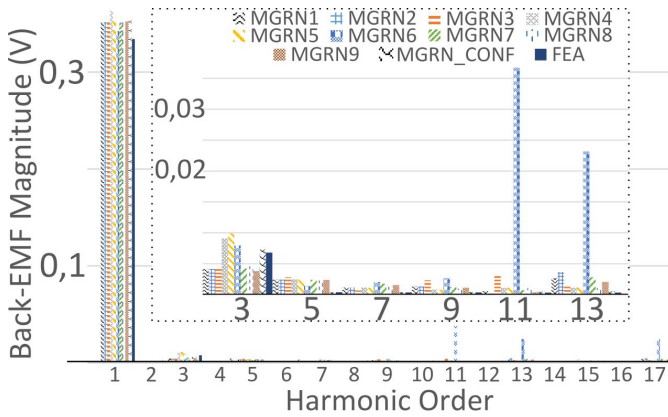


Fig. 10. Open-circuit back EMF harmonic content.

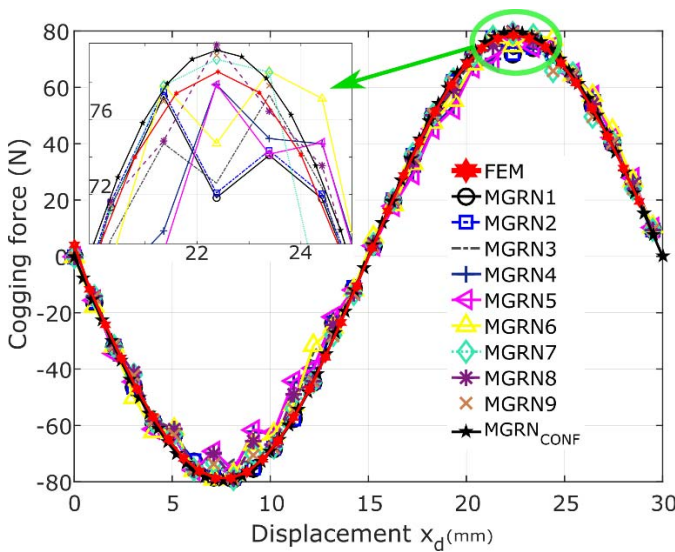


Fig. 11. Cogging force versus displacement.

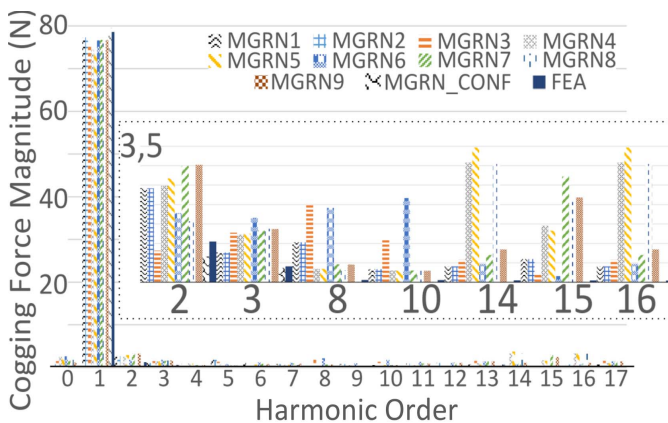


Fig. 12. Cogging force harmonic content.

Mesh number 5 shows the most significant difference in cogging force harmonic content when compared to FEA.

The conformal MGRN mesh pattern waveforms of open-circuit flux linkage, back EMF, and cogging force are the

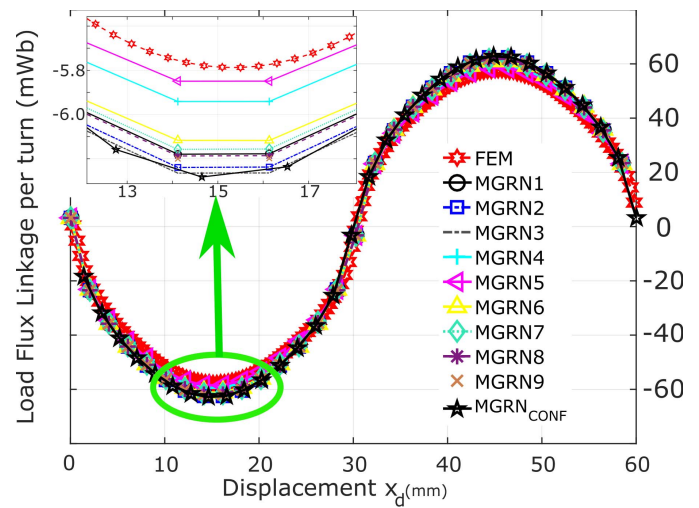


Fig. 13. Load flux linkage per turn versus displacement.

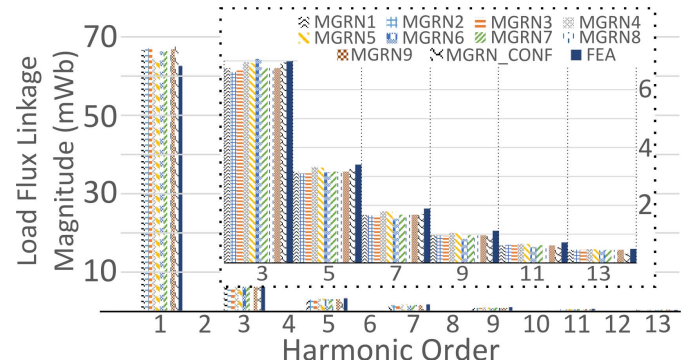


Fig. 14. Load flux linkage per turn harmonic content.

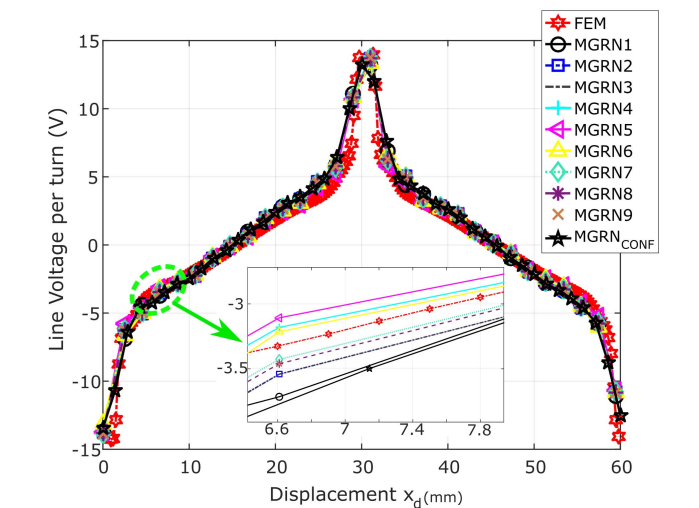


Fig. 15. Line voltage per turn versus displacement.

closest of all tested mesh patterns when comparing their harmonic content to those of the FEA model.

Figs. 13 and 14 illustrate the flux linkage per turn waveforms versus displacement and under load condition and its harmonic content, respectively. Figs. 15 and 16 show, respectively, the line voltage waveforms and its harmonic content

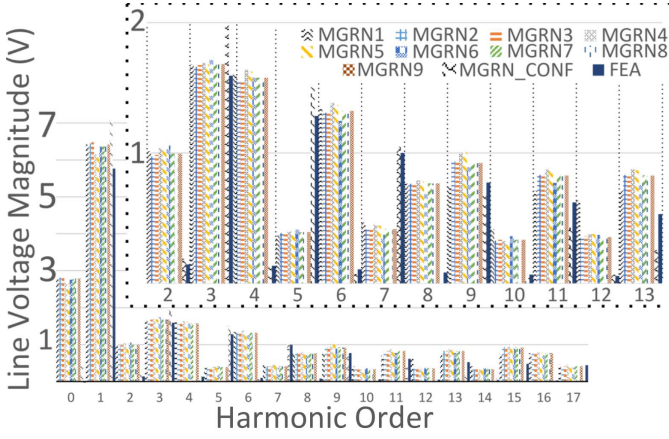


Fig. 16. Line voltage harmonic content.

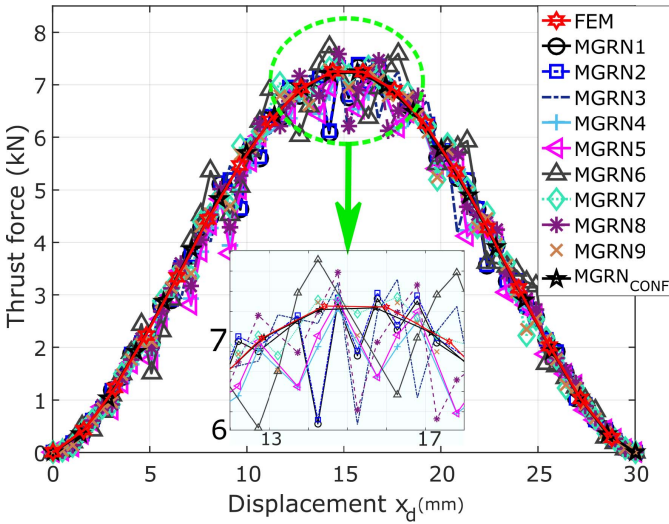


Fig. 17. Thrust force versus displacement.

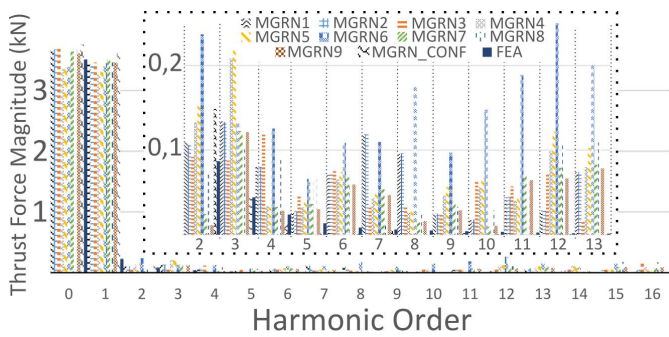


Fig. 18. Thrust force harmonic content.

under the same operating condition. Figs. 17 and 18 show the thrust force waveforms and harmonic content, respectively, for the 10 MGRN mesh patterns and the FEA model. The most harmonic content discrepancies are observed on MGRN mesh patterns 3–6, especially on thrust force (see Fig. 18).

A fairly good agreement is obtained between all nine non-conformal MGRN mesh schemes and FEA results under both open-circuit and on-load conditions. However, the closest waveforms, when comparing the harmonic content of

 TABLE III
MESH SCHEMES COMPARISON

Mesh	Number of nodes		Relative Error (%)							Computation Time (p.u)	
			Open-circuit			Load				Open-circuit	Load
			Φ_{Err}	V_{Err}	F_{Err}	Φ_{Err}	V_{Err}	F_{Err}			
FEA	3562	—	—	—	—	—	—	—	—	1	1
1	1716	1638	78	1.1	5.68	1.57	6.4	8.22	4.56	0.090	0.180
2	1566	1428	138	1.1	5.70	1.55	7.1	9.00	3.94	0.080	0.170
3	837	675	162	1.1	5.70	3.65	7.6	9.50	3.75	0.040	0.090
4	377	270	107	4.4	9.17	3.59	2.8	6.23	9.14	0.020	0.048
5	294	207	87	2.5	7.26	4.08	1.4	4.81	8.94	0.017	0.043
6	200	138	62	1.4	3.73	2.60	5.8	8.10	2.82	0.011	0.034
7	2581	2574	78	0.9	5.54	1.53	6.0	8.24	3.29	0.150	0.280
8	1452	1386	66	1.1	5.74	1.63	6.5	8.61	3.58	0.070	0.160
9	3264	3168	96	1.3	5.90	1.80	6.7	8.81	4.02	0.220	0.360
Conf.	3240	3240	0	1.6	6.60	0.58	7.6	9.15	1.03	0.230	0.437

all evaluated quantities (i.e., open-circuit and load flux linkage, EMF, and force), are given by the conformal MGRN mesh pattern model. Table III gives the number of central and interface nodes and the total number of nodes for different mesh patterns. In addition, the relative error on rms values of flux, voltage, and force and normalized computation time, under open-circuit and load conditions, is also given in Table III. The relative error on the previously mentioned quantities is calculated by (10)–(12). Φ_{Err} , V_{Err} , and F_{Err} are the relative error on flux linkage, voltage, and force, respectively. Φ_{rms}^{FEA} , Φ_{rms}^{MGRN} , V_{rms}^{FEA} , V_{rms}^{MGRN} , F_{rms}^{FEA} , and F_{rms}^{MGRN} are the flux linkage, voltage, and force evaluated by FEA and MGRN, respectively. It can be seen that all mesh schemes result in a relative error not exceeding 100% for both open-circuit and load conditions. Mesh 5 and Mesh 6 are the best compromise between computation time and accuracy since they have the lowest number of nodes

$$\Phi_{Err} = \frac{\Phi_{rms}^{FEA} - \Phi_{rms}^{MGRN}}{\Phi_{rms}^{FEA}} \times 100 \quad (10)$$

$$V_{Err} = \frac{V_{rms}^{FEA} - V_{rms}^{MGRN}}{V_{rms}^{FEA}} \times 100 \quad (11)$$

$$F_{Err} = \frac{F_{rms}^{FEA} - F_{rms}^{MGRN}}{F_{rms}^{FEA}} \times 100. \quad (12)$$

V. CONCLUSION

The methodology for scalar magnetic potential interpolation coupling for non-conformal meshing in MGRN was presented and nine different MGRN non-conformal mesh schemes were compared to one MGRN conformal mesh pattern and to FEA. The interpolation coupling allows for mesh relaxation on different parts of the model and renders motion processing completely independent of mesh discretization. The assessment of MGRN mesh patterns was conducted by comparison of rms values and harmonic content of flux linkage, voltage, and force evaluation under both open-circuit and load operating condition. With the FEA providing a reference for computation time and accuracy, it is shown that MGRN results in a considerable reduction in CPU time, while accuracy remains acceptable (the relative error remains under 100% difference with FEA on rms values of evaluated quantities).

When compared to FEA or to MGRN conformal meshing, MGRN mesh configurations containing a higher number of non-conformal interfaces induce the highest disparities in harmonic content of global quantities. The MGRN with scalar magnetic potential interpolation coupling is a contribution to the standardization of procedures for generic and sufficiently accurate RN models. The advantages of automatic MGRN lie in model development duration and computation time. It constitutes then a powerful tool for pre-design and optimization of electromagnetic devices. These aspects are currently investigated in the case of electrical machines.

REFERENCES

- [1] S. Asfirane, S. Hlioui, O. De La Barriere, Y. Amara, G. Barakat and M. Gabsi, "Software tool based on reluctance network automatic generation for electromagnetic devices modeling," in *Proc. NUMELEC*, 2017, pp. 1–3.
- [2] S. Asfirane, S. Hlioui, Y. Amara, O. D. La Barriere, G. Barakat, and M. Gabsi, "Global quantities computation using mesh-based generated reluctance networks," *IEEE Trans. Magn.*, vol. 54, no. 11, pp. 1–4, Nov. 2018, Art. no. 7002304.
- [3] D. Petrichenko, "Calculation and simulation of turbogenerators using permeance network," (in English), Optimization Application. Electric Power, Ecole Centrale de Lille, Villeneuve-d'Ascq, France, 2007.
- [4] *Electromagnetics Suite, Release 16.2*, ANSYS, Canonsburg, PA, USA, 2015.
- [5] T. J. E. Miller, M. McGilp, and A. Wearing, "Motor design optimisation using SPEED CAD software," in *Proc. IEEE Seminar Practical Electromagn. Des. Synthesis*, Feb. 1999, pp. 2/1–2/5.
- [6] B. D. Peloux, L. Gerbaud, F. Wurtz, V. Leconte, and F. Dorschner, "Automatic generation of sizing static models based on reluctance networks for the optimization of electromagnetic devices," *IEEE Trans. Magn.*, vol. 42, no. 4, pp. 715–718, Apr. 2006.
- [7] S. Ouagued, A. A. Diriye, Y. Amara, and G. Barakat, "Mesh relaxation effect on results quality/computation time ratio of a hybrid semi-numerical model," in *Proc. 11th Int. Conf. Ecological Vehicles Renew. Energies*, Apr. 2016, pp. 1–5.
- [8] H. Nguyen-Xuan, H. Dogan, S. Perez, L. Gerbaud, L. Garbuio, and F. Wurtz, "Efficient reluctance network formulation for electrical machine design using optimization," *IEEE Trans. Magn.*, vol. 50, no. 2, pp. 869–872, Feb. 2014, Art. no. 7021504.
- [9] R. Benlamine, T. Hamiti, F. Vangraefschèpe, F. Dubas, and D. Lhotellier, "Modeling of a coaxial magnetic gear equipped with surface mounted PMS using nonlinear adaptive magnetic equivalent circuits," in *Proc. IEEE 22nd Int. Conf. Electr. Mach. (ICEM)*, Lausanne, Switzerland, pp. 1888–1894, Sep. 2016.
- [10] M. A. Benhamida, H. Ennassiri, Y. Amara, G. Barakat, and N. Debbah, "Study of switching flux permanent magnet machines using interpolation based reluctance network model," in *Proc. Int. Conf. Elect. Sci. Technol. Maghreb (CISTEM)*, Marrakech, Morocco, 2016, pp. 1–7.
- [11] S. Asfirane *et al.*, "Performance comparison of different winding configurations of a novel wound-field flux-switching linear machine," in *Proc. IEEE 13th Int. Conf. Electr. Machines (ICEM)*, Alexandroupoli, Greece, Sep. 2018, pp. 2276–2282.